AMATH 403 HW 5

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May 15, 2026

Question 1: 4.3.18. Use separation of variables to solve the Helmholtz boundary value problem $\Delta u = u$, $u(x,0) = 0$, $u(x,1) = f(x)$, $u(0,y) = 0$, $u(1,y) = 0$, on the unit square $0 < x, y < 1$.

Using separation of variables:

$$u(x,y) = v(x)w(y)$$

Helmholtz equation:

$$\Delta u = u$$
$$u_{xx} + u_{yy} = u$$

Using substitution:

$$u_{xx} = v''(x)w(y)$$
$$u_{yy} = v(x)w''(y)$$
$$v''(x)w(y) + v(x)w''(y) = v(x)w(y)$$

Divide by $v(x)w(y)$

$$\frac{v''(x)}{v(x)} + \frac{w''(y)}{w(y)} = 1$$
$$\frac{v''(x)}{v(x)} = 1 - \frac{w''(y)}{w(y)} = \lambda$$

Finding the first ODE:

$$v''(x) = \lambda v(x)$$
$$v''(x) - \lambda v(x) = 0$$

Finding the second ODE:

$$\frac{-w''(y)}{w(y)} = \lambda - 1$$
$$-w''(y) = (\lambda - 1)(w(y))$$
$$w''(y) - (1-\lambda)w(y) = 0$$

Based on the given boundary conditions

$$u(0,y),\ v(0)w(y) = 0 \implies v(0) = 0$$
$$u(1,y),\ v(1)w(y) = 0 \implies v(1) = 0$$

Solving $v''(x) - v(x) = 0$

Case $\lambda = 0$:

$$\begin{gathered} v''(x) = 0 \\ v(x) = Ax + B \\ v(0) = B = 0 \\ v(1) = A(1) + 0 = 0 \\ v(x) = 0 \end{gathered}$$

Case 2 $\lambda > 0,\ \lambda = k^2$:

$$\begin{gathered} v''(x) - k^2 v(x) = 0 \\ v(x) = Ae^{kx} + Be^{-kx} \\ v(0) = Ae^0 + Be^0 = 0 \\ v(1) = Ae^k + Be^{-k} = 0 \end{gathered}$$

Case $\lambda < 0,\ \lambda = -k^2$

$$\begin{gathered} v''(x) + k^2 v(x) = 0 \\ v(x) = A\cos(kx) + B\sin(kx) \\ v(0) = A(1) + B(0) = 0 \implies A = 0 \\ v(1) = (0)\cos(k) + B\sin(k) = 0 \\ \sin(k) = 0 \implies k = n\pi,\ n = 1, 2, 3, \ldots \end{gathered}$$

Solving $w(y)$:

$$\begin{gathered} \lambda_n = -(n\pi)^2 \\ w''(y) - (1 - \lambda_n)w(y) = 0 \\ w''(y) - (1 + n^2\pi^2)w(y) = 0 \end{gathered}$$

$$\begin{gathered} \mu_n = \sqrt{1 + n^2\pi^2} \\ w_n(y) = C_n \cosh(\mu_n y) + D_n \sinh(\mu_n y) \\ u(x, 0) = 0 \\ w(0) = C_n \cosh(0) + D_n \sinh(0) = 0 \\ C_n(1) + D_n(0) \implies C_n = 0 \\ W_n(y) = D_n \sinh(y\sqrt{1 + n^2\pi^2}) \end{gathered}$$

By superposition:

$$u(x, y) = \sum_{n=1}^{\infty} b_n \sin(n) \sinh(y\sqrt{1 + n^2\pi^2})$$

Finding b_n:

$$u(x,1) = f(x)$$

$$f(x) = \sum_{n=1}^{\infty} b_n \sin(n) \sinh(y\sqrt{1+n^2\pi^2})$$

$$b_n \sinh(y\sqrt{1+n^2\pi^2}) = 2\int_0^1 f(x)\sin(nx\pi)\,dx$$

$$b_n = \frac{2}{\sinh(y\sqrt{1+n^2\pi^2})}\int_0^1 f(x)\sin(nx\pi)\,dx$$

Question 2: Give an example of functions f and g, along with an explicit solution u, such that u attains its maximum value at an interior point of the domain D.

$$-\Delta u = f$$

In the unit open disk

$$D := \{(x, y) : x^2 + y^2 < 1\}$$

Dirichlet boundary on the unit circle C which is the boundary

$$u = g,\ \partial D = C = \{(x, y) : x^2 + y^2 < 1\}$$

Use a function that has its peak at the interior:

$$1 - x^2 - y^2$$

Evaluate at boundary:

$$\begin{aligned} u(x, y) &= 1 - (x^2 + y^2) \\ &= 1 - 1 \\ &= 0 \\ g(x, y) &= 0 \end{aligned}$$

$$\begin{aligned} \Delta u &= f \\ -(u_{xx} + y_{yy}) &= f \\ -(-2 + -2) &= f \\ f(x, y) &= 4 \end{aligned}$$

$$\begin{aligned} u(x, y) &= 1 - x^2 - y^2 \\ f(x, y) &= 4 \\ g(x, y) &= 0 \end{aligned}$$

Question 3: Textbook Problem 7.1.5

(a) Prove $\hat{\delta}(k) = \frac{1}{\sqrt{2\pi}}$

$$\hat{f}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x)\, e^{-ikx}\, dx$$
$$\hat{\delta}(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x)\, e^{-ikx}\, dx$$

$$f(x) = e^{-ikx}$$
$$\int_{-\infty}^{\infty} \delta(x)\, f(x)\, dx$$

Using the sifting property:

$$\int_{-\infty}^{\infty} \delta(x)\, f(x)\, dx = f(0)$$
$$\int_{-\infty}^{\infty} \delta(x)\, e^{-ikx}\, dx = f(0)$$
$$f(0) = e^{-ik0} = e^{0} = 1$$

Thus

$$\delta(\hat{k}) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \delta(x)\, f(x)\, dx = \frac{1}{\sqrt{2\pi}}$$
$$\delta(\hat{k}) = \frac{1}{\sqrt{2\pi}}(1) = \frac{1}{\sqrt{2\pi}}$$

(b) Use the Fourier inversion formula and part (a) to find the Fourier transform of $e^{i\omega x}, \omega \in \mathbb{R}$

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx}\, dk$$

$$\text{We know } f(x) = \delta(x),\ \hat{f}(k) = \frac{1}{\sqrt{2\pi}}$$

$$\delta(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{ikx}\, dk$$

$$\delta(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ikx}\, dk$$

Change Variables

$$\delta(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{iku}\, dk$$

$$\delta(u) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{ixu}\, dx$$

$$\int_{-\infty}^{\infty} e^{iux}\, dx = 2\pi\delta(u)$$